THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 9–11: **Life Processes in Plants, Light: Shadows and Reflections, Earth, Moon and the Sun** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Plants that make their own food are called: 1

(a) decomposers **(b)** heterotrophs **(c)** autotrophs **(d)** parasites

2. Light travels in: 1

(a) straight lines **(b)** circles **(c)** zigzag lines **(d)** curved lines

3. The Earth spins on its axis from: 1

(a) east to west **(b)** west to east **(c)** south to north **(d)** north to south

4. Starch stored in leaves turns iodine solution: 1

(a) yellow **(b)** colourless **(c)** red **(d)** blue-black

5. For a shadow to form, we need: 1

(a) a source of light, an opaque object and a screen
(b) a screen only
(c) an object only
(d) a source only

6. The longest day in the northern hemisphere is on about: 1

(a) 23 September **(b)** 22 December **(c)** 21 March **(d)** 21 June

7. Rhizobium bacteria live in the root nodules of: 1

(a) cereals **(b)** leguminous plants
(c) grasses **(d)** palms

8. The bouncing back of light from a surface is: 1

(a) dispersion **(b)** reflection **(c)** refraction **(d)** absorption

9. The Moon revolves around the Earth in about: 1

(a) 365 days **(b)** 27 days **(c)** 12 hours **(d)** 1 day

10. Plants are called the producers because they: 1

(a) make food for all **(b)** are green only
(c) eat others **(d)** decompose waste

11. The image formed by a plane mirror is: 1

(a) virtual and inverted **(b)** virtual and erect
(c) real and inverted **(d)** real and erect

12. A solar eclipse occurs when the ___ comes between the Sun and Earth. 1

(a) Moon **(b)** Venus **(c)** Mars **(d)** Earth

13. The process by which green plants make food using sunlight is: 1

(a) digestion **(b)** transpiration **(c)** respiration **(d)** photosynthesis

14. Objects that give out their own light are: 1

(a) luminous objects **(b)** non-luminous objects

(c) opaque objects only **(d)** transparent objects

15. One rotation of the Earth takes about: 1

(a) 12 hours **(b)** 24 hours **(c)** 30 days **(d)** 365 days

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. What is photosynthesis? Write the word equation. 2

17. What is the difference between luminous and non-luminous objects? 2

18. What are rotation and revolution of the Earth? 2

19. Differentiate between xylem and phloem. 2

20. What is a pinhole camera? 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. Write the word equation for photosynthesis and explain each term. 3

22. Explain with an activity that light travels in straight lines. 3

23. Explain how day and night occur on the Earth. 3

24. Explain how a pitcher plant gets food. 3

OR

A person stands 2 m in front of a plane mirror. Find the distance of the image from the mirror and between the person and the image. 3

25. How does a lunar eclipse occur? 3

OR

Why do farmers grow leguminous crops in rotation? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe photosynthesis with the raw materials, conditions, products and the place where it occurs. 5

OR

Describe the activity to show that light travels in a straight line and to produce a shadow, and discuss the factors affecting the size of a shadow. 5

27. Explain the rotation and revolution of the Earth with the consequences of each. 5

OR

Describe how plants respire and excrete. 5

END OF QUESTION PAPER **Total: 50 Marks**

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